

## Quiz 7

**Instructions:** There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

**Problem 1.** (6 points)

(a) Use the Chain Rule to find  $\frac{\partial z}{\partial s}$  and  $\frac{\partial z}{\partial t}$  when  $z = e^{x+2y}$  with  $x = s/t$  and  $y = t/s$ .

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = e^{x+2y} \left( \frac{1}{t} \right) + 2e^{x+2y} \left( -\frac{t}{s^2} \right) \\ &= e^{x+2y} \left( \frac{1}{t} - \frac{2t}{s^2} \right) \\ &= e^{\frac{s}{t} + \frac{2t}{s}} \left( \frac{1}{t} - \frac{2t}{s^2} \right)\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = e^{x+2y} \left( -\frac{s}{t^2} \right) + 2e^{x+2y} \left( \frac{1}{s} \right) \\ &= e^{x+2y} \left( -\frac{s}{t^2} + \frac{2}{s} \right) \\ &= e^{\frac{s}{t} + \frac{2t}{s}} \left( -\frac{s}{t^2} + \frac{2}{s} \right)\end{aligned}$$

(b) Use the implicit differentiation to find  $\frac{dy}{dx}$  at the point  $(1, 2)$ , where

$$x^2 + xy + y^2 - 7 = 0.$$

$$\text{let } F(x, y) = x^2 + xy + y^2 - 7$$

$$F_x = 2x + y, \quad F_y = x + 2y$$

$$\Rightarrow F_x(1, 2) = 2 + 2 = 4$$

$$F_y(1, 2) = 1 + 4 = 5$$

$$\text{thus } \frac{dy}{dx} \Big|_{(1,2)} = -\frac{F_x}{F_y} = -\frac{4}{5}$$

**Problem 2.** (4 points) Given a function  $f(x, y) = \sin(xy)$ . Find the directional derivative of  $f$  at the point  $(1, 0)$  along the direction of the vector  $\mathbf{v} = \langle 1, 2 \rangle$ , and in what direction does  $f$  increase the most rapidly at the point  $(1, 0)$  and what is the corresponding rate of change?

$$\nabla f(x, y) = \langle y \overset{f_x}{G_x}(x, y), x \overset{f_y}{G_y}(x, y) \rangle$$

$$\Rightarrow \nabla f(1, 0) = \langle 0, 1 \rangle$$

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\langle 1, 2 \rangle}{\sqrt{1^2 + 2^2}} = \langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \rangle$$

$$\begin{aligned} \Rightarrow D_{\mathbf{u}} f|_{(1,0)} &= \nabla f(1, 0) \cdot \mathbf{u} = \langle 0, 1 \rangle \cdot \langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \rangle \\ &= \frac{2}{\sqrt{5}}. \end{aligned}$$

$f$  increases most rapidly in the direction of  $\nabla f(1, 0) = \langle 0, 1 \rangle$  and the corresponding rate of change is  $|\nabla f(1, 0)| = |\langle 0, 1 \rangle| = 1$ .